Do structural instability measures predict sector ETF relationship breakdown?

Executive research briefing

This reproducible study uses the original eleven ETFs and the reference's historical nine pairs. Its 5542 daily observations reproduce the reported sample dates and 70/30 boundaries. See baseline replication for selection and numerical discrepancies; exact original-input replication is not claimed.

Can instability be measured? Yes, trailing OLS coefficient departures, residual dispersion, AR persistence and valid monotonic half-lives can be measured using only observed history. These are descriptive indicators, not validated declarations of discrete market regimes.

Does instability predict deterioration, and add to simple predictors? 0 of 4 designated primary comparisons show positive forecast-loss improvement with Holm $p<0.05$, a positive block-bootstrap lower confidence limit, and the expected negative convergence association in development and holdout. The primary analysis does not establish credible incremental predictive value. Detectable historical instability alone is insufficient evidence of useful forecasting or trading gains. The baseline forecast already includes spread magnitude, recent volatility, recent spread changes and matured convergence. Statistical associations alone are not evidence of useful incremental forecasting.

Exploratory results are mixed: 2 BY-adjusted comparisons favor the extended forecast, while 32 favor the simpler baseline. The rolling 504-day beta_instability / 20-day future-phi improvement is exploratory evidence for persistence forecasting, requiring prospective replication and economic validation. The static 504-day beta_instability / 60-day convergence improvement has greater convergence at higher indicator values, contrary to the proposed deterioration direction. These exploratory findings do not overturn the primary result.

Are results credible? The design separates development, validation and test, purges crossing labels, groups dependence by shared dates, and adjusts for multiplicity. Nevertheless, the final sample has few independent 60-day blocks, rare entries and possible longer persistence. Treat results as conditional historical evidence requiring prospective replication.

Does acting on instability improve performance? Final-holdout comparisons:

- static/beta_entry: net return 2.00% versus 2.00%, Sharpe 0.258 versus 0.258, utilization 70.2% versus 70.2% - static/beta_reduction: net return 1.72% versus 2.00%, Sharpe 0.233 versus 0.258, utilization 67.9% versus 70.2% - static/vol_entry: net return 2.00% versus 2.00%, Sharpe 0.258 versus 0.258, utilization 70.2% versus 70.2% -

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Executive briefing, continued

rolling/beta_entry: net return 1.62% versus 1.56%, Sharpe 0.328 versus 0.312, utilization 40.3% versus 42.1% - rolling/beta_reduction: net return 1.43% versus 1.56%, Sharpe 0.296 versus 0.312, utilization 40.2% versus 42.1% - rolling/vol_entry: net return 1.56% versus 1.56%, Sharpe 0.312 versus 0.312, utilization 42.1% versus 42.1%

Changes must also be compared with volatility filtering and the validation-scaled control. Lower capital utilization can reduce drawdowns without adding information. Costs from exposure changes are included; daily rebalancing cost sensitivities are in the full report.

How sensitive are conclusions? Across 36 holdout convergence specifications, date-weighted loss gains range from -0.0824 to 0.2173. At full-family BY 5%, 1 favor the extended forecast and 5 favor the simpler baseline. Validation and test agree on the sign of loss improvement in 26 of 36 specifications. These are exploratory robustness counts, not independent confirmations. Calendar-three-year mean, boundary carry and executable turnover costs are also reported. No best threshold or specification is selected. Negative and insufficient-sample results remain in `predictive_results.csv`.

Read the research report for tables, figures and explicit limits.

Historical nine-pair baseline replication

pair	beta	sigma	static entries	static completed trades	static return	static sharpe	rolling return	rolling sharpe
XLU/XLK	0.7081	0.0807	1	0	-0.2933	-0.3529	0.1789	0.3036
XLU/XLI	0.8326	0.0856	4	3	0.2423	0.5356	0.2595	0.5793
XLB/XLU	0.8663	0.0926	5	4	0.3243	0.5414	0.2496	0.5169
XLV/XLY	0.8906	0.0797	7	7	0.5085	0.8564	0.1812	0.3766
XLB/XLP	0.7253	0.1053	2	1	-0.0227	0.0120	0.1427	0.3593
XLI/XLV	0.8653	0.0872	3	2	0.0303	0.0993	-0.0468	-0.0678
XLU/XLY	0.6539	0.1040	5	4	0.4243	0.6505	0.3422	0.6405
XLB/XLI	0.7530	0.0680	3	3	0.1689	0.5024	-0.0253	-0.0528
XLB/XLV	0.6438	0.1075	3	2	0.1193	0.2291	0.2020	0.4375

Reference-log convention: normalized log-spread changes compounded as daily returns. Entries include open trades; completed trades are separate. Returns use decimal fractions. Full signed, absolute and relative target differences are in the baseline assessment; exact original-input replication is not certified.

Primary incremental forecast comparisons | Holdout

strategy	feature	n	n dates	mse gain	gain ci low	gain ci high	p gain	q primary gain holm
rolling	beta_instability	1839.0000	479.0000	0.0060	-0.0063	0.0195	0.3928	0.6950
rolling	phi	1839.0000	479.0000	0.0111	-0.0021	0.0269	0.1145	0.4578
static	beta_instability	3309.0000	479.0000	-0.0028	-0.0081	0.0023	0.3295	0.6950
static	phi	3309.0000	479.0000	0.0066	-0.0043	0.0184	0.2317	0.6950

252-session features, 20-session convergence outcomes. Positive loss gain favors instability. Bootstrap uses 60-session shared-date blocks, conditional on development models. Holm corrects four primary comparisons. Judge expected direction, confidence intervals, multiplicity and the small number of independent blocks together; consult the executive findings for the conclusion.

Net economic comparison | Final holdout

strategy	variant	return	sharpe	max drawdown	volatility	capital utilization	completed trades	transaction costs
static	control	0.0200	0.2583	-0.0646	0.0421	0.7016	8	0.0008
rolling	control	0.0156	0.3118	-0.0349	0.0261	0.4206	6	0.0008
static	beta_entry	0.0200	0.2583	-0.0646	0.0421	0.7016	8	0.0008
static	beta_reduction	0.0172	0.2334	-0.0682	0.0405	0.6792	8	0.0009
static	vol_entry	0.0200	0.2583	-0.0646	0.0421	0.7016	8	0.0008
static	risk_scaled_control	0.0200	0.2583	-0.0646	0.0421	0.7016	8	0.0008
rolling	beta_entry	0.0162	0.3278	-0.0343	0.0257	0.4026	6	0.0008
rolling	beta_reduction	0.0143	0.2956	-0.0360	0.0253	0.4019	6	0.0009
rolling	vol_entry	0.0156	0.3118	-0.0349	0.0261	0.4206	6	0.0008
rolling	risk_scaled_control	0.0156	0.3118	-0.0349	0.0261	0.4206	6	0.0008

Simple adjusted leg returns, original scalar-position cost convention. Entry filters preserve existing exits; reduction halves gross exposure. Other pairs receive no compensating capital. Daily leg-rebalancing costs and alternative windows are separately reported. Consult the full Markdown report for assumptions and all negative/insufficient specifications.


XLB/XLI | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
OLS beta
126 days
252 days
504 days
Frozen beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date


XLB/XLP | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
126 days
252 days
504 days
Frozen beta
OLS beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date

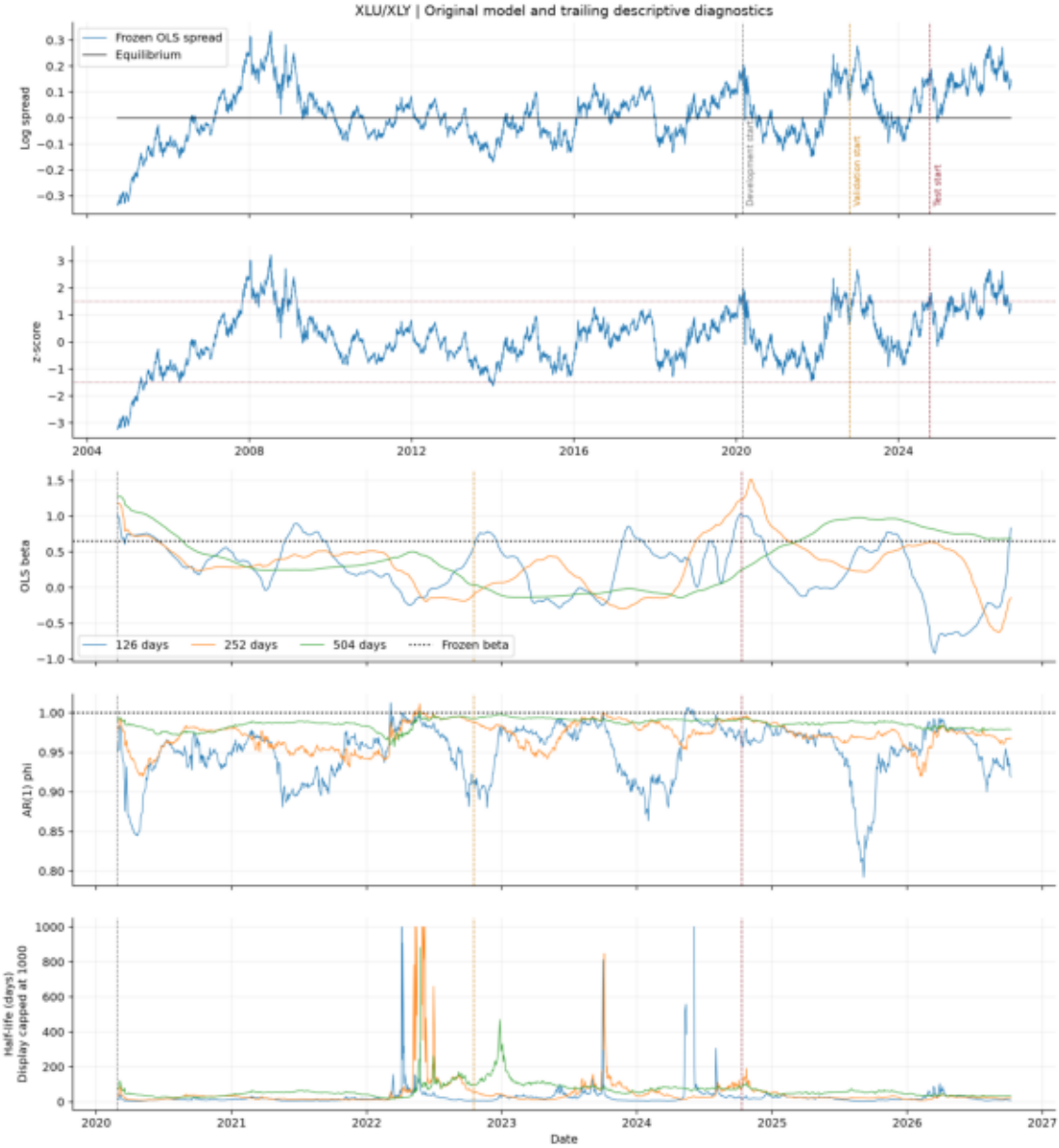

XLB/XLU | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
126 days
252 days
504 days
Frozen beta
OLS beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date


XLB/XLV | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
126 days
252 days
504 days
Frozen beta
OLS beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date

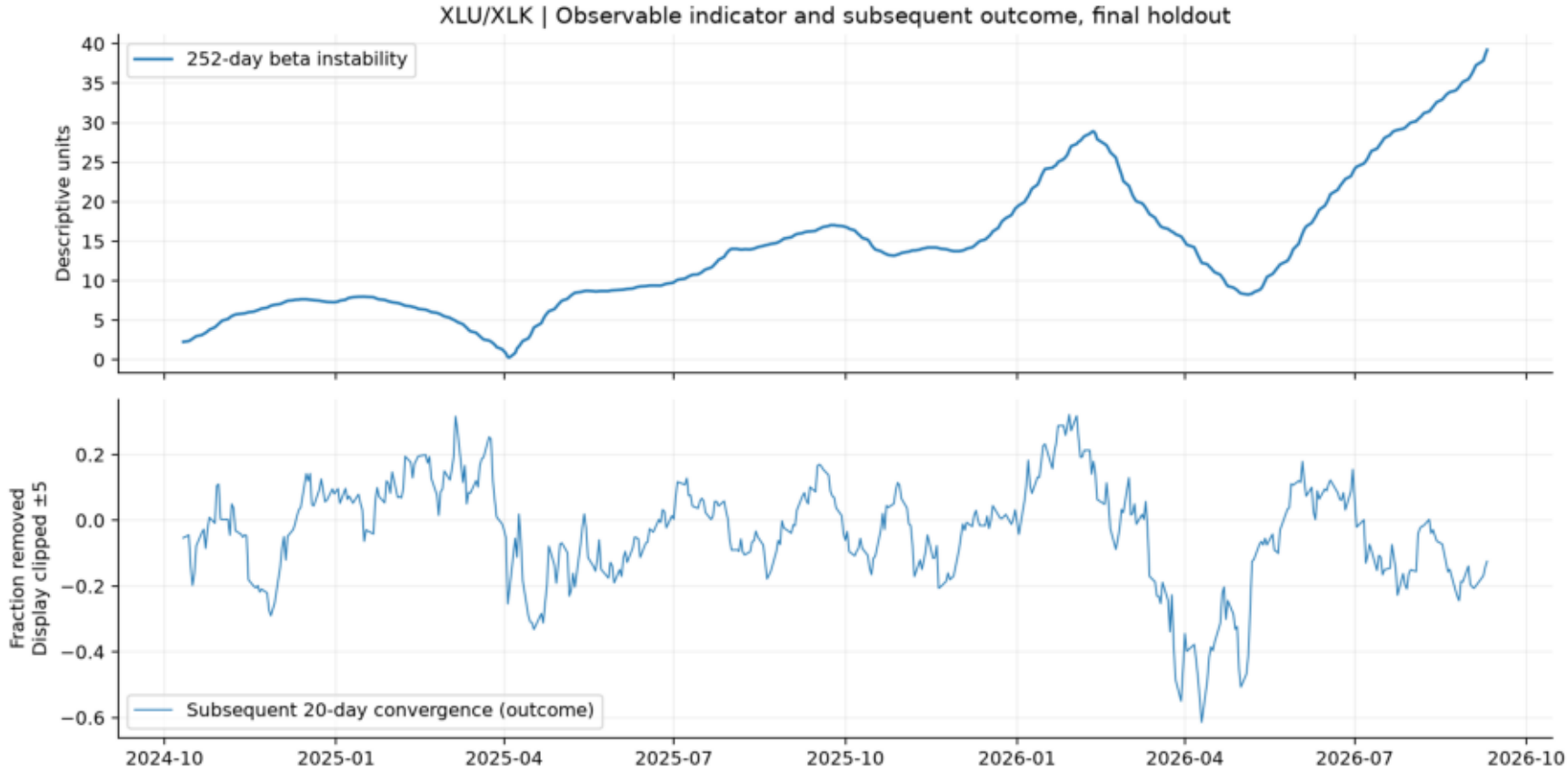

XLI/XLV | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
126 days
252 days
504 days
Frozen beta
OLS beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date

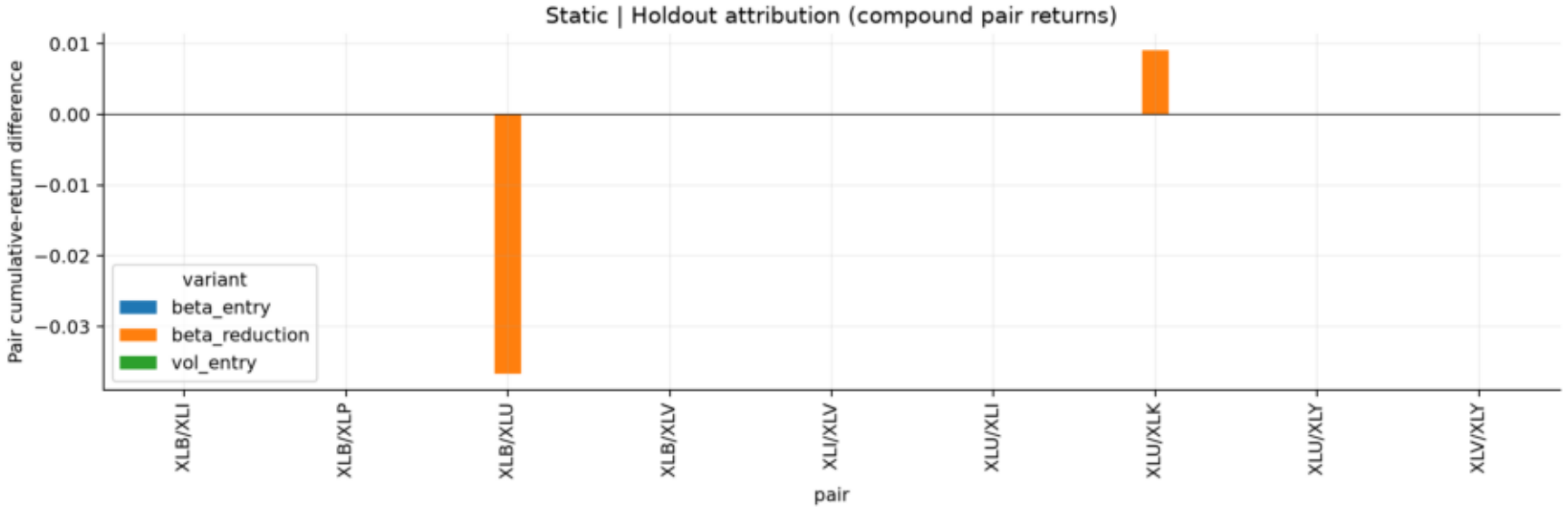

XLU/XLI | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
126 days
252 days
504 days
Frozen beta
OLS beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date

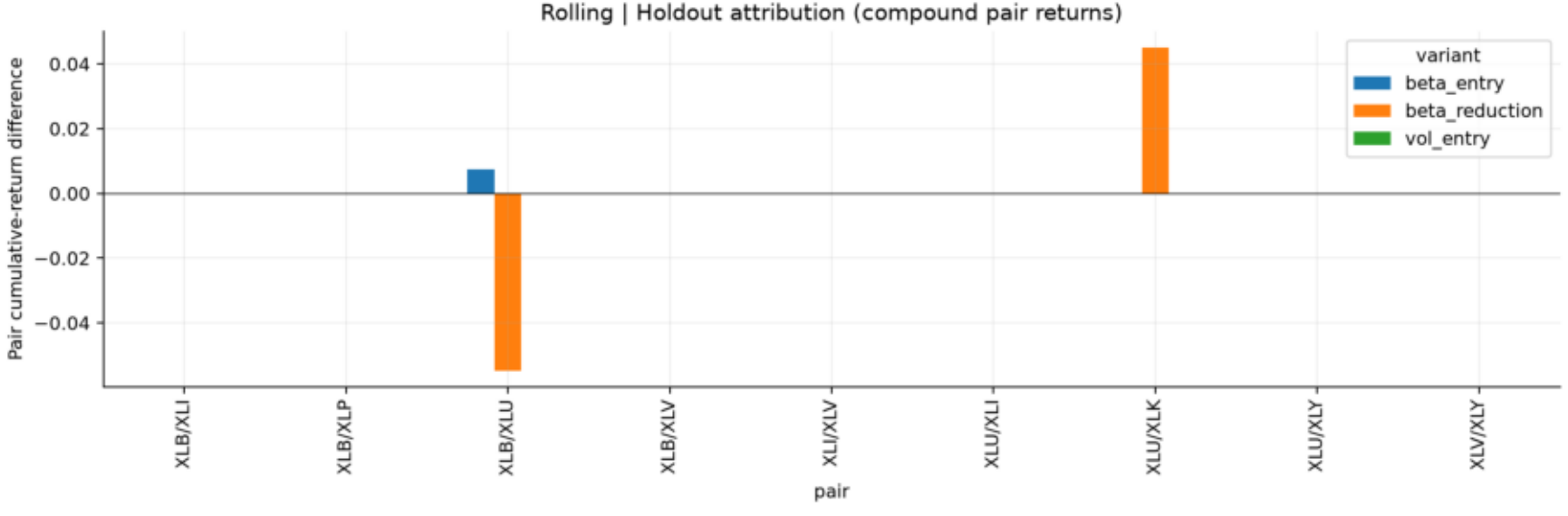

XLU/XLK | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
126 days
252 days
504 days
Frozen beta
OLS beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date

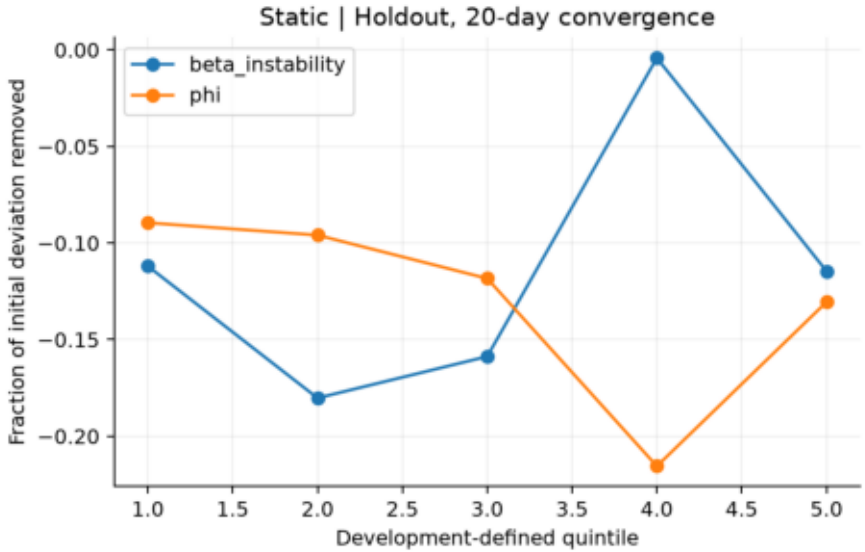

XLU/XLY | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development start
Validation start
Test start
z-score
OLS beta
126 days
252 days
504 days
Frozen beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date

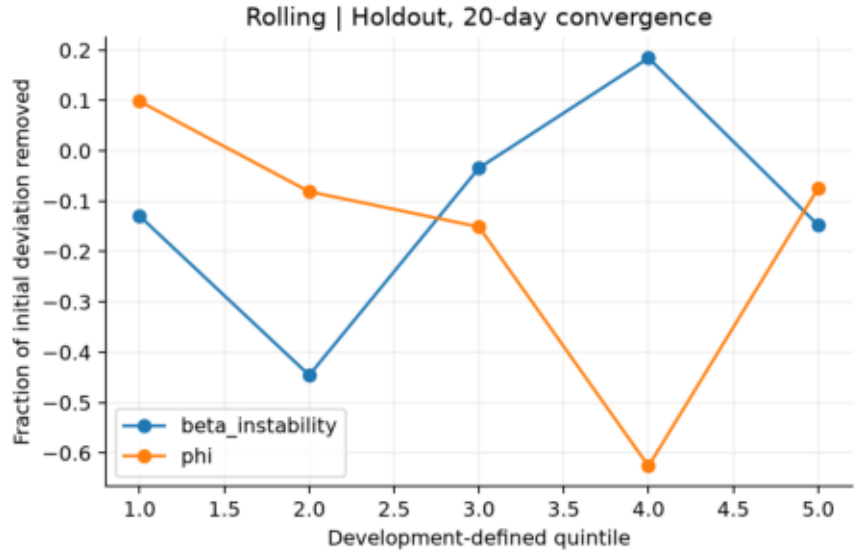

XLV/XLY | Original model and trailing descriptive diagnostics
Frozen OLS spread
Equilibrium
Log spread
Development end
Validation start
Test start
z-score
126 days
252 days
504 days
Frozen beta
OLS beta
AR(1) phi
Half-life (days)
Display capped at 1000
Date

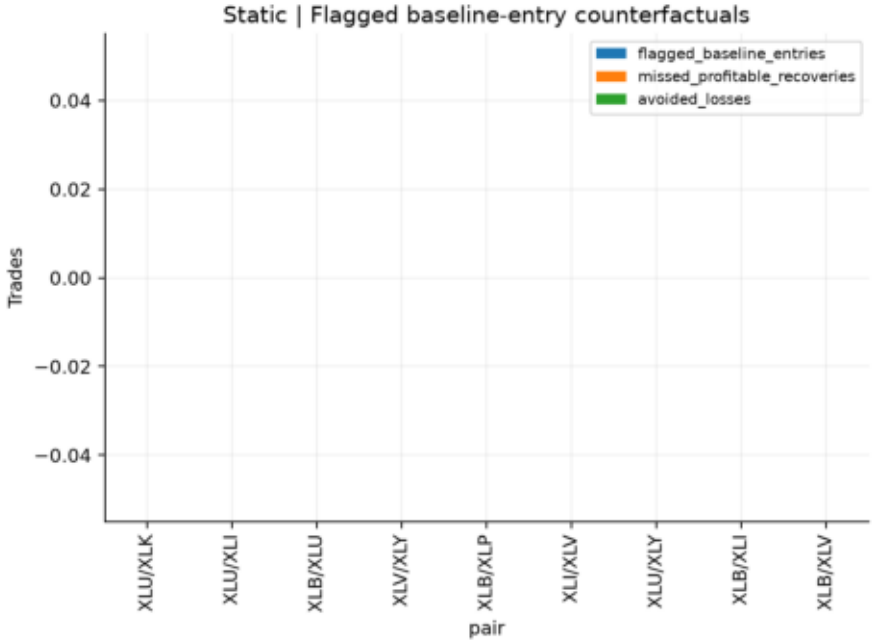

XLU/XLK | Observable indicator and subsequent outcome, final holdout
252-day beta instability
Descriptive units
40
35
30
25
20
15
10
5
0
Fraction removed
Display clipped ±5
0.2
0.0
−0.2
−0.4
−0.6
Subsequent 20-day convergence (outcome)
2024-10
2025-01
2025-04
2025-07
2025-10
2026-01
2026-04
2026-07
2026-10

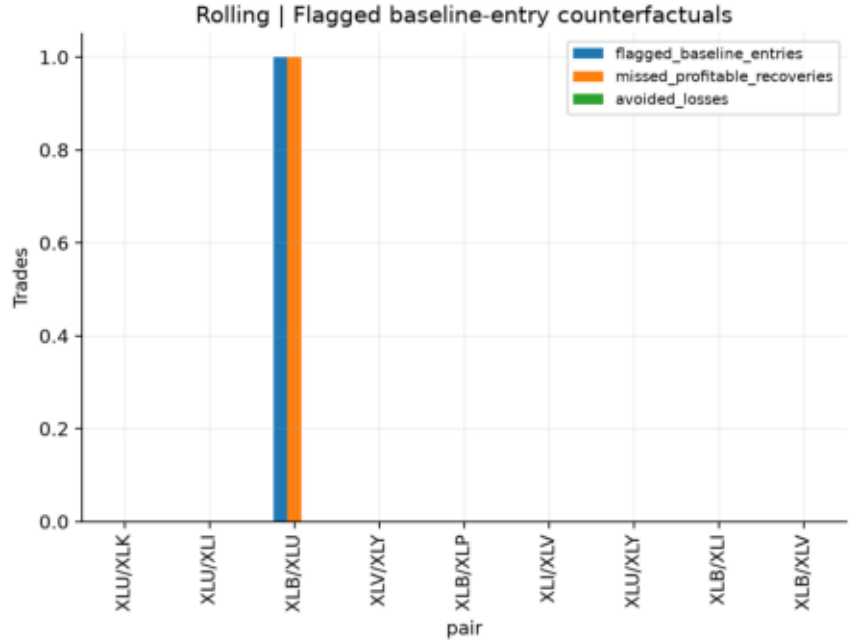

Static | Holdout attribution (compound pair returns)
Pair cumulative-return difference
0.01
0.00
−0.01
−0.02
−0.03
variant
beta_entry
beta_reduction
vol_entry
XLB/XLI
XLB/XLP
XLB/XLU
XLB/XLV
XLI/XLV
XLU/XLI
XLU/XLK
XLU/XLY
XLV/XLY
pair


Rolling | Holdout attribution (compound pair returns)
Pair cumulative-return difference
0.04
0.02
0.00
−0.02
−0.04
variant
beta_entry
beta_reduction
vol_entry
XLB/XLI
XLB/XLP
XLB/XLU
XLB/XLV
XLI/XLV
XLU/XLI
XLU/XLK
XLU/XLY
XLV/XLY
pair


Static | Holdout, 20-day convergence
beta_instability
phi
Fraction of initial deviation removed
0.00
−0.05
−0.10
−0.15
−0.20
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Development-defined quintile


Rolling | Holdout, 20-day convergence
beta_instability
phi
Fraction of initial deviation removed
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Development-defined quintile


Static | Flagged baseline-entry counterfactuals
flagged_baseline_entries
missed_profitable_recoveries
avoided_losses
Trades
0.04
0.02
0.00
−0.02
−0.04
XLU/XLK
XLU/XLI
XLB/XLU
XLV/XLY
XLB/XLP
XLI/XLV
XLU/XLY
XLB/XLI
XLB/XLV
pair


Rolling | Flagged baseline-entry counterfactuals
flagged_baseline_entries
missed_profitable_recoveries
avoided_losses
Trades
1.0
0.8
0.6
0.4
0.2
0.0
XLU/XLK
XLU/XLI
XLB/XLU
XLV/XLY
XLB/XLP
XLI/XLV
XLU/XLY
XLB/XLI
XLB/XLV
pair


Static mean | Out-of-sample portfolio, net of reference costs
control
beta entry
beta reduction
vol entry
risk scaled control (test only)
Growth of $1
1.20
1.15
1.10
1.05
1.00
Development start
Validation start
Test start
Rolling mean | Out-of-sample portfolio, net of reference costs
Growth of $1
1.15
1.10
1.05
1.00
control
beta entry
beta reduction
vol entry
risk scaled control (test only)
2020
2021
2022
2023
2024
2025
2026
2027